ACUMENUS

ZEPHYRUS · HOSPITAL OPERATIONS INTELLIGENCE

# Hummingbird

## Consolidated Implementation Plan

*The native mobile companion to Zephyrus — Android · iOS · shared Kotlin Multiplatform core*

**ACUM-ENG-HMB-001**

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**CONFIDENTIAL**

Native mobile companion to Zephyrus. The web app answers “what is happening across the hospital, and why?” on wall displays and workstations. This document is the single authoritative plan for the phone: it states the cross-cutting **invariants** once (Part II), sequences the work as six self-contained **phases** (Part III), and backs both with a full parity matrix and reference appendices (Part IV).

> **Status** Planning v1 · 2026-06-26. Bounded by the locked authentication rules in `.claude/rules/auth-system.md` — all backend work described here is strictly *additive*. Deep-dive docs (00–07) and the seven-stream code audit (`research/`) provide field-level detail.

# Contents

PART I

# Strategy & Context

## 1. Executive summary

Hummingbird is the **glance-and-act** surface of Zephyrus. The web app answers "what is happening across the hospital, and why?" on wall displays and workstations. Hummingbird answers a sharper question for a worker mid-shift, in a hand, in three seconds:

> "What needs **me**, right now — and can **I** do it in two taps?"

One native app per platform adapts its home, its unified **"For You"** action queue, and its notification defaults to the signed-in worker — from the **transporter** moving a patient to the **CMO** glancing at house status between meetings.

### The headline decisions

| Decision | Choice | Rationale |
|---|---|---|
| **UI** | Native per platform (Compose + SwiftUI) | Glanceability and feel are the product; native widgets, Live Activities, Critical Alerts |
| **Shared code** | **KMP** domain + data core, bridged via SKIE | Clinical/status logic is "defensible" — it must not drift between two codebases |
| **Backend access** | A versioned **mobile BFF** (`/api/mobile/v1/*`) | The web `/api` is chatty, inconsistently authed, PHI-rich |
| **Auth** | **Sanctum tokens** (+ OIDC/PKCE path), *additive* | Move off session cookies without touching the locked flow |
| **Real-time** | **Push-first**, websocket-when-foregrounded, poll fallback | Battery + correctness; never hold a background socket |
| **Offline** | Offline-*aware* (cache reads, queue non-critical writes) | Spotty hospital Wi-Fi; block safety-critical writes offline |
| **Notifications** | Server-side **four-tier "earned-urgency" router** | Alarm fatigue is a clinical-safety failure, not a UX nit |
| **Design** | One **DTCG token source** → Compose + SwiftUI + CSS | One design system, three platforms, zero drift |

### The plan in one picture

```
P0 Foundation   ######          auth · push · BFF · tokens · RBAC · assignment model
P1 Frontline      ######        Transport · EVS · RTDC beds/barriers · For-You · notif v1
P2 Decisions        ########    Ops approvals · Command/Brief · OR board · huddles
P3 Awareness            ######  ED signals · Staffing · transfers · watch/widgets
P4 Net-new                ######## PI/PDSA · ED actions · integration health
P5 Hardening/GA               ###### security + clinical-safety review · perf · rollout
```

*~9–12 months to GA for a focused team; usable internal pilots from end of P1. Phases P1–P2 can overlap once the BFF pattern is set.*

### The sequencing logic (why this order)

Three platform gaps gate *everything*, so they come first (P0). Then we build on **live data with low backend risk** for fast value — RTDC, Ops approvals, Transport, EVS all run on existing endpoints (P1–P2). Features that require **net-new backend** (ED clinician actions, Process

Improvement) are deferred (P4) because their web surfaces are mock/stub today; we still ship their *glanceable* signals earlier where the data already exists (P3). Hardening and the safety/security gates that GA legally depends on close the program (P5).

## 2. Why a companion, not a port

Zephyrus is a dense, instrument-rich **operations bridge** built for multi-panel study. Most of that does not belong on a phone under time pressure. Every web capability is triaged against the glance-and-act question into one treatment:

| Flag | Meaning | Mobile treatment |
|---|---|---|
| **GLANCE** | Read at a glance; situational awareness | Home tiles, widgets, Live Activities |
| **ACT** | A worker would *act* on it from a phone | First-class flows + notification actions |
| **NOTIFY** | A discrete event worth a push | The earned-urgency taxonomy |
| **WEB** | Too analytical/dense for phone | Deep-link to web; not rebuilt |

**Rule of thumb** If a task needs more than ~3 visible panels, a keyboard, or sustained study (utilization analytics, process mining, simulation authoring, blueprint modeling, admin), it **stays on the web** and Hummingbird offers a one-tap deep link.

This plan honors the Zephyrus canon on mobile: **Rigorous · Composed · Defensible**; **earned urgency** (color and pushes are rationed); **status never by color alone**; **dark-default**, `tabular-nums`, gold focus; **replace the EHR, don't reskin it**. It is not a shrunk dashboard, not a consumer health app, and never an alarm-fatigue pager.

## 3. What the audit found

A seven-stream parallel audit (95 models, ~106 API endpoints, all four workflows + supporting domains + platform) produced five findings that shape the entire sequence.

### Per-domain readiness (this drives phasing)

| Domain | Data model | API contract | Live data | Real-time | Verdict |
|---|---|---|---|---|---|
| **RTDC** | ✓ rich | ✓ `/api/rtdc/*` | ✓ event-sourced | ✓ Reverb WS | Flagship — build first |
| **Ops / AI approvals** | ✓ `ops.*` | ✓ `/api/ops/*` | ✓ production-grade | — | Reuse endpoints verbatim |
| **Transport / EVS / Staffing** | ✓ | ✓ endpoints | ✓ event logs | — | Textbook mobile workers |
| **Perioperative** | ✓ rich | ✓ `/api/cases,blocks` | △ mostly mock UI | ✗ local timers | Ready data, no surface |
| **Command Center** | ✓ | ✓ `/api/command-center/*` | ✓ | — | Strain index + Exec Brief |
| **Emergency Dept** | △ 2 tables | ✗ 0 ED endpoints | △ via Command/Analytics | ✗ | Glance now, act later (NEW) |
| **Process Improvement** | △ models only | ✗ stub | ✗ empty stubs | ✗ | Net-new backend |

### The three platform gaps that gate everything

1. **No mobile auth.** Session-cookie only; **Sanctum is installed but dormant** (`User` lacks `HasApiTokens`, zero `auth:sanctum` routes). OIDC/Authentik exists but also lands a web session. → token auth is **net-new** (additive).
2. **No push infrastructure.** Zero APNs/FCM, no device registry, no notifier. Only Reverb websockets (foreground) + one demo SSE. → push is **net-new**.
3. **Inconsistent API auth & shape.** Several operational endpoints (`cases`, `blocks`, `rooms`, `providers`, analytics, `improvement/*`) carry **no `auth` middleware**; no uniform envelope; PHI interleaved with operational data. → a **BFF** is required.

Full detail: `research/` (`02-rtdc`, `03-perioperative`, `04-improvement-and-ops`, `05-platform-auth-realtime-design`, `06-supporting-domains-and-api`, `01-emergency-department`, `07-mobile-best-practices`).

## 4. Personas — "all levels of the delivery system"

One app, **role-adaptive home**. Zephyrus treats its audience as one product switched by a `workflow_preference` (a *preference*, not a hard permission); Hummingbird honors that and extends down to the frontline mobile worker the web under-serves.

| # | Persona | Altitude | Phase | Home emphasis |
|---|---|---|---|---|
| P1 | Transporter / Porter | Frontline worker | 1 | My trips + available jobs |
| P2 | EVS Technician | Frontline worker | 1 | My bed-turns + next dirty bed |
| P3 | Bedside / Charge Nurse | Frontline unit | 1–2 | My unit: census, barriers, incoming |
| P4 | OR Circulating / Charge Nurse | Frontline OR | 2 | Live room board + my cases |
| P5 | Nursing Supervisor / Bed Manager | Ops leader | 1 | House bed-need + pending requests |
| P6 | Patient-Flow / Capacity Lead | Ops leader | 2 | Capacity vs demand + approvals inbox |
| P7 | Perioperative Manager | Ops leader | 2–3 | OR day at a glance |
| P8 | PI / Quality Lead | Ops leader | 4 | My PDSA cycles + barrier trends |
| P9 | Executive (CMO/COO/CNO) | Executive | 2 | One quiet strain index + Brief |
| P10 | Staffing Coordinator | Ops leader | 3 | Open requests + below-safe units |

### The altitude model

How "earned urgency" expresses per level: a frontline worker's home is a **task list**; a unit's home is **my unit / my cases**; an ops leader's home is **the house / my domain**; an executive's home is a **single strain index** that stays quiet until it shouldn't. Full per-persona journeys: `reference/02-core-functionality-by-role.md`.

PART II

# The Invariants

These decisions are constant across all phases. They are stated once here; each phase references them rather than re-deciding.

## 5. Architecture

**Native UI per platform over a shared KMP domain/data core.** This single decision must be made **now** — committing to shared Kotlin logic puts Kotlin/Native in the iOS build, and deciding late forces an expensive re-architecture. If the org rejects KMP, the floor is two fully-separate apps bound by the API contract + a shared conformance test spec.

```
PRESENTATION (native)   Android: Compose + ViewModel (MVVM-UDF; MVI for dense boards)
                        iOS:     SwiftUI + @Observable (TCA only for true state machines)
                        → widgets · Live Activities/Dynamic Island · Watch/Wear

SHARED (KMP)            domain   models · use-cases · status rules · urgency scoring ·
validation
                        data     repositories · Ktor BFF client · SQLDelight cache · outbox ·
                                 sync engine · Reverb/Pusher realtime · token store
                        tokens   generated from DTCG (see §8)

PLATFORM (expect/actual) Keychain/Keystore · biometrics · APNs/FCM · BGTasks/WorkManager
```

- **Android:** Compose (Material3 themed to Zephyrus tokens, *not* default Material), Hilt, Coroutines/Flow, Ktor, SQLDelight, DataStore, WorkManager, FCM, Glance widgets, Wear.
- **iOS:** SwiftUI + Observation, Swift Concurrency, Ktor-via-shared (or URLSession actual), SQLDelight, Keychain, BGTaskScheduler, APNs (+ Critical Alerts, ActivityKit), WidgetKit, watchOS. **SKIE** generates idiomatic Swift over the KMP module.
- **Feature modules** are vertical slices depending only on `shared/domain` + `shared/data` + `core-ui`. The **For-You** feature composes every domain through one `ForYouFeed` use-case in `shared/domain` — the single home of the earned-urgency ranking.

Module/repo structure and the full stack rationale: `reference/03-architecture.md`.

## 6. Real-time & offline model

**Three-mode real-time** (mirrors the web's Reverb posture and extends it for battery):

| App state | Transport | Behavior |
|---|---|---|
| **Foreground, screen on** | **Reverb WS** (Pusher protocol) | Live tiles; **re-snapshot all visible queries on (re)connect** — Reverb does not replay |
| **Background / locked** | **Push (APNs/FCM)** | Silent push → WorkManager/BGTask delta-sync; visible push for NOTIFY events |
| **No connectivity** | Poll on resume + cache | Stale badge; reconcile outbox |

We **never hold a websocket in the background**. Public WS channels are **PHI-free** (counts/ids only); any future PHI-on-wire needs `PrivateChannel` + token channel auth.

### Offline-aware, not offline-first

- **Reads:** cache-then-network; emit cached immediately with an **as-of timestamp**, refresh, re-emit; on failure keep cache + show a **stale badge** (never a blank screen).
- **Non-critical writes** (acknowledge, claim routine job, routine status): optimistic local apply + **outbox**; flush on reconnect.
- **Safety-critical writes** (approve an action, place a bed, complete an isolation bed-turn, advance a case past a clinical gate): **require connectivity**; if offline, the action is **disabled with an explicit reason** — never silently queued.
- **Conflicts:** last-write-wins + field-merge + server `version`/ETag; **409** surfaces “changed since you loaded — review,” never a blind overwrite. No CRDTs.

## 7. The Mobile BFF & auth model

*A versioned BFF (`/api/mobile/v1/`) is the only surface the apps talk to.** It reuses existing models/services and **delegates all mutations** to the existing lifecycle services (`OperationalActionLifecycleService`, the RTDC engine, the Transport/EVS/Staffing `*OperationsService`s) — it reshapes, it does not re-implement.

### BFF rules

- **One screen → one call** where practical; **uniform envelope** `{ data, meta:{as_of,stale,version}, links:{web} }`.
- **`auth:sanctum` on everything**; **role-scoped** and **PHI-minimized** (tokenized/initials in lists; full PHI only on an explicit authorized detail call, **never** in lists or notifications).
- ETag/cursor/`updated_since` for concurrency and delta sync.

### Auth (additive — never touches the locked flow)

Add `HasApiTokens` to `User`; issue **Sanctum personal access tokens** with role-matched abilities; provide an **OIDC+PKCE** path reusing Authentik for SSO orgs. The token endpoints **honor `must_change_password`** by returning a scoped change-challenge (the temp-password/Resend/superuser flow is preserved verbatim). Tokens live in Keychain/Keystore gated by biometrics; short-lived access + rotating refresh; idle auto-lock; server-side revoke for remote wipe.

### RBAC

Reads stay broad (house status is for everyone at their altitude, PHI-gated); **writes** get Laravel **Policies** so a device can only act within role/assignment (only the assigned transporter progresses a trip; only an authorized approver decides an action). This requires the **assignment model** (Phase 0). The contract: `api-contract/hummingbird-bff.v1.yaml` (OpenAPI 3.1, validated). Full spec: `reference/04-backend-requirements.md`.

## 8. Design-system parity

**One DTCG token source → three platforms, verified non-divergent in CI.** The web keeps its Tailwind setup; `design-tokens/tokens.json` generates Compose + SwiftUI equivalents via Style Dictionary and a CI job **diffs** resolved values against `tailwind.config.js` + `.impeccable/design.json` — a mismatch fails the build.

The tokens encode the Zephyrus canon: the **Two-System Rule** (operational blue/slate `#2563EB`/`#3B82F6` vs. brand crimson `#9B1B30` + gold `#C9A227` focus-only), the **rationed status ramp** (teal `#2DD4BF` / amber `#E5A84B` / coral `#E85A6B` / sky `#60A5FA` dark), **Figtree weights 400/500/600 only**, dark-default, the 4px grid. Signature components (`KpiTile` with its 3px status stripe + arrow + label, `StatusChip` dot-not-color-alone, `Panel` quiet-lift) are built **once per platform** from these tokens. Pipeline + samples: `design-tokens/`.

## 9. Earned-urgency notifications

A notification is the most expensive signal Hummingbird can send. The taxonomy is a **clinical-safety component** with acceptance criteria — a mis-tuned one causes alarm fatigue (staff silence the app and miss the real breach). **Four tiers:**

| Tier | Meaning | iOS | Android | Quiet hours |
|---|---|---|---|---|
| **T1 Critical breach** | Threshold breach / safety event, act *now* | Critical Alert (entitlement) | Full-screen intent | Overrides (capped) |
| **T2 Actionable** | Assigned to you to act on soon | Time-Sensitive + actions | High-importance + actions | Deferred unless STAT |
| **T3 Awareness** | Worth knowing, not urgent | Passive | Default/low | Silenced |
| **T4 Digest** | Periodic summary | Scheduled | Scheduled low | Silenced |

### Principles

- Severity earned per event (most state changes are GLANCE, silent).
- **Route to the person who can act** (needs the assignment model).
- The **action is in the notification** (Approve/Claim/Acknowledge from the lock screen, zero app-open).
- Respect the human (per-tier opt-in, **quiet hours**, **per-shift budget**, dedupe/coalesce).
- **No PHI ever** (generic copy, CI payload linter).
- **Escalate, don't repeat** (unacked T1/T2 → re-deliver → next responsible → web command center, fully logged).

The complete event → tier → audience → action taxonomy: `reference/05-notifications-earned-urgency.md`.

## 10. Security & HIPAA gates

Security is a **release gate**, not a backlog. The cheapest, highest-impact wins, done first: **no PHI in notifications/logs/app-switcher snapshots**, **biometric-gated token storage + auto-lock**, **TLS + cert pinning with no PHI on public channels**.

Plus: encrypted local cache (SQLCipher), `FLAG_SECURE`/privacy-view, jailbreak/root + attestation (App Attest/Play Integrity), MDM/remote-wipe, action-level RBAC, tamper-evident audit logging, BAA-covered or PHI-free SDKs only, and WCAG 2.2 AA (status-never-by-color, dynamic type, VoiceOver/TalkBack, 44pt targets). The seven non-negotiable **release-gate criteria** are in `reference/06-security-hipaa.md §9` and are written into the program Definition of Done.

## 11. The shared core every worker gets

Regardless of role/phase, every persona's app is built from the same primitives — only the **feed sources and ranking weights** change:

1. **Secure entry** — token login, biometric unlock, auto-lock, forced-change honored.
2. **Role-aware home** — the altitude's most important truth, glanceable; workflow switcher.
3. **The "For You" queue** — one prioritized, cross-domain list of what needs *this* worker, ranked by earned-urgency score, each card carrying its **primary action inline**.
4. **Rationed, actionable notifications** (§9).
5. **House/unit status glance** — role-scoped, PHI-gated; tile + widget + (phased) Live Activity/complication.
6. **Directory & search** — the web's ⌘K as a search screen.
7. **Profile/preferences & deep links** — default workflow, notification tiers, quiet hours, theme; "Open in Zephyrus" for every WEB-deferred path.

This is why one app spans the porter and the CMO: shared primitives, role-tuned feed.

PART III

# The Phased Plan

Each phase is a self-contained work package: **thesis · why now · backend · mobile · features & personas · notifications · exit criteria**. All phases assume the Invariants (Part II).

## Phase 0 — Foundation

~6–8 WEEKS · THE UNBLOCK

### Thesis

A signed-in worker on both platforms can authenticate with biometrics, hit a secured BFF, receive a push, and see one real RTDC tile. Nothing user-facing ships, but everything after depends on it.

### Why now

The three gating gaps (§3) block 100% of features. Building them first is the only rational sequence.

### Backend

- Sanctum token auth: `HasApiTokens` on `User`; `POST /api/auth/token{,/refresh,/revoke}` + `/auth/change-password` — additive, honoring `must_change_password`. OIDC+PKCE path stubbed.
- `mobile_devices` registry + push service (APNs `.p8` HTTP/2 + FCM v1) behind a `PushNotifier` contract; queued via Horizon.
- **BFF scaffold** (`/api/mobile/v1/*`): uniform envelope, role-scoping, PHI-minimization, `/me`, `/realtime/config`, `/rtdc/census`.
- **Assignment model** — `user_id` FKs on owners (today `Barrier.owner`/`RtdcPlan.owner` are free-text) + a `user_unit` pivot. *Prerequisite for both the For-You queue and notification routing.*
- RBAC Policy layer; **remediate the auth-less operational endpoints** (a standing security issue regardless of mobile).

### Mobile (both platforms)

- Repo + CI scaffold; **KMP `shared` skeleton** (`domain`/`data`/`platform`); SKIE bridge.
- **Design-token pipeline** wired (DTCG → Compose/SwiftUI); `core-ui` primitives (`KpiTile`, `StatusChip`, `Panel`). *(Already started — `design-tokens/`.)*
- Auth flow: token + **biometric unlock** + auto-lock + Keychain/Keystore; cert pinning.
- BFF client (Ktor), SQLDelight cache, outbox skeleton, push registration.

### Features / personas

None user-facing (foundation).

### Notifications

Delivery pipeline stood up; one test push.

> **Exit criteria** Auth regression suite green (locked flow unchanged); a device receives a test push; one live `/rtdc/census` tile renders on iOS *and* Android via the BFF; security checklist §1–§3 pass.

## Phase 1 — Frontline workers & bed flow

~6–8 WEEKS · HIGHEST, FASTEST VALUE

### Thesis

Transporters and EVS techs run their jobs from the phone; bed managers place beds and clear barriers on the move; everyone gets a working **For-You** queue and rationed pushes.

### Why now

All **LIVE** endpoints, clean lifecycles, lowest backend risk, and the clearest mobile win (transporter/EVS *are* mobile users; bed managers want placement off-workstation).

### Backend

- BFF: RTDC (`/rtdc/census`, `/house`, `/bed-requests` + `/{id}/decision`, `/barriers` + `/{id}/resolve`), Transport (`/queue`, `/{id}/status|assign|handoff`), EVS (`/queue`, `/{id}/status`).
- Add **`BarrierUpdated`** broadcast (barriers don't broadcast today, unlike census/huddle).
- **Server-side evaluators** for events with no native trigger: SLA breach (`needed_at`/`at_risk`), capacity deficit (`bed_need>0`, available→0), barrier aging, unplaced bed request. Wire into the **notification router v1** (tiers, routing via assignment model, quiet hours, budget).

### Mobile

- **Transport (P1):** claim → run trip (17-status) → structured handoff; STAT push.
- **EVS (P2):** claim → start (SOP/PPE for isolation) → complete (notifies bed manager).
- **RTDC beds & barriers (P3/P5):** live unit census + safe capacity (Reverb WS); pending bed requests + **transparent placement decision** (accept/edit/reject); log/resolve barriers.
- **For-You queue v1** across these domains; first **widget / Live Activity** (active trip; unit/house status glance).

### Features / personas

P1 Transporter, P2 EVS, P3 Charge Nurse (beds/barriers), P5 Bed Manager — *activated*.

### Notifications

STAT transport/isolation bed-turn assigned (T1); new pending bed request, unplaced bed, aging barrier, SLA at-risk (T2); bed-turn completed (T3).

> **Exit criteria** A transporter claims a STAT job from the lock screen in **< 10 s**; a bed manager places a bed from a push; evaluators fire correctly **within the per-shift budget**.

## Phase 2 — Decisions & the OR board

~8–10 WEEKS · THE MARQUEE OPS-LEADER + OR PHASE

### Thesis

Ops leaders approve and assign operational actions on the go; executives get a single quiet strain screen + Brief; OR staff get a live board, case tracking, and safety-note SLAs.

### Why now

Ops approvals reuse `/api/ops/*` **verbatim** (highest value, near-zero backend), and the OR data/contract already exist (needs only three bug-fixes). This is the decision-making heart of the product.

### Backend

- BFF: Ops (`/ops/inbox`, `/approvals/{id}/decision`, `/actions/{id}/{assign,start,complete}`) delegating to `OperationalActionLifecycleService`; Command Center (`/command/house`, `/brief`); OR (`/or/board`, `/cases/today`, `/cases/{id}`, `/{id}/status`, safety-notes, milestones, transport).
- **Fix the OR backend bugs first:** `ORCaseController@store` writes string `status` not `status_id`; analytics SQL references non-existent `or_cases.actual_start/end_time` (route via `orlog`+`case_metrics`); reference endpoints filter `is_active` but the column is `active_status`.
- Add the **huddle action-items API** (`RtdcPlan` table exists with no API/UI today).

### Mobile

- **Ops approvals (P6):** approve/reject/assign/start/complete + agent-inbox glance tiles.
- **Command Center / Exec Brief (P9):** strain index (0–4), hero KPIs with trust badges, the server-composed brief; sparse exec notifications.
- **Perioperative (P4/P7):** live room board + my cases + single-case tracking; advance status; safety-note SLA acknowledge; pre-op milestone ack; transport-to-OR.
- **RTDC huddles (P3/P6):** unit-huddle steps (WS-synced) + huddle action-items.
- Dynamic Island / Live Activity for room board + house strain; **iOS Critical Alerts + Android full-screen-intent wired for T1 only** (apply for entitlements now — they gate GA).

### Features / personas

P4 OR Nurse, P6 Capacity Lead, P7 Periop Manager, P9 Executive — *activated*; P3 gains huddles.

### Notifications

Recommendation awaiting approval, action expiring/overdue, safety-note SLA, pre-op milestone incomplete (T2; Crit safety-note + house-status escalation T1); case delayed, first-case late start, cancellation (T2/T3).

> **Exit criteria** A capacity lead approves an action from a notification; an OR nurse acknowledges an overdue safety note; the exec home is a single quiet strain screen that escalates correctly.

## Phase 3 — Awareness breadth

~6–8 WEEKS

### Thesis

The glanceable signals that already exist as data — ED boarding/diversion/surge, staffing gaps — reach the right leaders; at-a-glance coverage extends to watch and widgets.

### Why now

These are **GLANCE/NOTIFY** built on existing computed metrics; they broaden situational awareness without the net-new backend that ED *actions* and PI need.

### Backend

- BFF: ED signals (`/ed/signals` — boarding, diversion, LWBS, surge from Command/Analytics); Staffing (`/staffing/requests`, `/{id}/status`); regional/inter-facility transfer.
- **Diversion endpoint + `DiversionUpdated` broadcast** (model exists, no API/broadcast today).
- Evaluators: ED boarding band (≥6), ESI 1–2 LWBS, surge; staffing below-minimum-safe.

### Mobile

- **ED glanceable signals (P9/P5):** boarding, diversion, LWBS, surge tiles; ESI 1–2 LWBS as T1.
- **Staffing (P10):** open requests, below-safe alerts, create/source/assign/fill.
- **Watch / Wear** complications for house/unit status; widget breadth.

### Features / personas

P10 Staffing Coordinator — *activated*; P5/P9 gain ED awareness.

### Notifications

ED diversion active, ESI 1–2 LWBS, unit below minimum-safe (T1); ED boarding band crossing, boarder dwell >4h, surge critical, staffing unfilled/escalated (T2).

> **Exit criteria** ED breaches page the right leaders within budget; a staffing coordinator fills a gap from mobile; glance surfaces cover house status across widgets/watch.

## Phase 4 — Net-new backend features

~8–10 WEEKS · BACKEND-LED

### Thesis

The domains that are mock/stub on the web today get real backend, then their mobile-worthy slices.

### Why now

These require **net-new backend** (Process Improvement returns empty stubs; PDSA write routes referenced by the web don't exist; ED has no clinician write path). Doing them last lets the high-value live-data features ship first and de-risks the net-new work.

### Backend

- Build the missing **PI/PDSA** write paths; **ED boarder-dwell** evaluator and (if scoped) a net-new ED clinician write path; **integration-health** alerts (connector-down, dead-letter spike); simulation **read** (results/promoted recommendations — authoring stays web).

### Mobile

- **PI/PDSA (P8):** cycle ownership + stage advance + barrier ack.
- **ED actions:** boarder-dwell breaches; live-board/triage acknowledgements *if* the write path exists.
- **Integration health** glance + alerts for ops/admin.

### Features / personas

P8 PI/Quality Lead — *activated*; ED gains limited actions.

### Notifications

PDSA stage due, barrier assigned (T2); boarder dwell >4h (T2); connector down / dead-letter spike (T2/T3, ops/admin).

> **Exit criteria** PI leads manage PDSA on mobile; ED dwell breaches notify; ops sees feed health.

## Phase 5 — Hardening & GA

~4–6 WEEKS

### Thesis

The app passes its safety, security, performance, and accessibility gates and rolls out behind flags.

### Work

- Full **security + clinical-safety reviews** (notification-taxonomy sign-off); penetration test; **PHI-leak audit**; accessibility audit (WCAG 2.2 AA).
- Performance/battery tuning (WS lifecycle, background sync, BFF payload sizes).
- Offline-edge hardening (409/conflict UX, outbox reconciliation, stale-badge correctness).
- Store readiness (privacy nutrition labels, **Critical Alerts entitlement** approval, data-safety form); pilot → phased rollout behind feature flags.

> **Exit criteria** All release-gate criteria pass; pilot units validate time-to-act and notification

budgets; **GA**.

PART IV

# Reference

## Appendix A — Full feature-parity matrix

Maps **every** web data element/function to a mobile treatment, satisfying “map to all data elements and functionality.” `GLANCE / ACT / NOTIFY / WEB` · backend `LIVE / MOCK / NEW / WS` · phase. Field-level detail per domain: `reference/01-feature-parity-matrix.md` and `research/`.

### RTDC — Real-Time Demand & Capacity (flagship)

| Capability | Treatment | Backend | Phase |
|---|---|---|---|
| Unit census + acuity-adjusted safe capacity | GLANCE·NOTIFY | LIVE·WS | 1 |
| Hospital bed-need roll-up (net/deficit) | GLANCE·NOTIFY | LIVE·WS | 1 |
| Bed (status/isolation/gender) | GLANCE | LIVE | 1 |
| Bed request — list / **create** | GLANCE·ACT·NOTIFY | LIVE | 1 |
| Bed placement decision — **accept/edit/reject** (transparent score) | ACT | LIVE | 1 |
| Barrier to discharge — list / **log / resolve** | GLANCE·ACT·NOTIFY | LIVE | 1 |
| Huddle (unit/service/global) + steps | GLANCE·ACT | LIVE·WS | 2 |
| Huddle **action items** (owner/due) | ACT·NOTIFY | NEW | 2 |
| Discharge prediction / reconciliation / GMLOS | GLANCE | LIVE | 2 |
| Care-journey milestone | GLANCE | LIVE | 2 |
| Diversion event | GLANCE·NOTIFY | NEW | 3 |
| Discharge readiness / priorities | GLANCE·ACT | NEW (mock) | 3 |
| RTDC deep analytics / trends | WEB | MOCK | — |

### Perioperative (ready data, greenfield surface)

| Capability | Treatment | Backend | Phase |
|---|---|---|---|
| OR case (service/surgeon/room/status) | GLANCE·ACT·NOTIFY | LIVE | 2 |
| Wheels clock (ORLog, 13 timestamps) | GLANCE | LIVE | 2 |
| Case status — **advance** (Sched→InProg→Delay→Comp) | ACT·NOTIFY | LIVE (fix store) | 2 |
| Room status board (derived) | GLANCE·NOTIFY | LIVE | 2 |
| Case timing (progress %/variance) | GLANCE·NOTIFY | LIVE | 2 |
| Safety note — **create/ack** (SLA 15/30/60/120m) | GLANCE·ACT·NOTIFY | LIVE | 2 |
| Pre-op milestone — **ack** (H&P/Consent/Labs) | ACT·NOTIFY | LIVE | 2 |
| Case transport — ready/**complete** | ACT·NOTIFY | LIVE | 2 |
| Provider directory / reference data | GLANCE / lookup | LIVE | 2 |
| Block/primetime/turnover/provider/service analytics; forecasts | WEB | LIVE/MOCK | — |

## Ops / AI & Command Center (“approvals on the go”)

| Capability | Treatment | Backend | Phase |
|---|---|---|---|
| Recommendation (title/risk/rationale) | GLANCE·NOTIFY | LIVE | 2 |
| Operational action lifecycle (draft → … → completed, expires +8h) | GLANCE·ACT·NOTIFY | LIVE | 2 |
| **Approve / reject** approval ★ | ACT·NOTIFY | LIVE | 2 |
| Assign / start / complete (override/expire P3) | ACT | LIVE | 2/3 |
| Agent inbox summary | GLANCE | LIVE | 2 |
| House strain/surge index (0–4) + hero KPIs + 24h forecast | GLANCE·NOTIFY | LIVE | 2 |
| Executive Brief (situation/plan/impact/confidence) | GLANCE·NOTIFY | LIVE | 2 |
| Intervention/metric trust/lineage/freshness | GLANCE | LIVE | 3 |
| Data-quality finding (stale feed) | GLANCE·NOTIFY | LIVE | 3 |
| Simulation author / ops graph / process mining | WEB | LIVE | — |

## Emergency Department (glance now, act later)

| Capability | Treatment | Backend | Phase |
|---|---|---|---|
| ED boarding count (≤6) · LWBS % · door-to-provider/LOS · surge | GLANCE·NOTIFY | LIVE (Command/Analytics) | 3 |
| Active diversion · ESI 1–2 LWBS (safety) | GLANCE·NOTIFY | LIVE/NEW | 3 |
| Boarder dwell >4h since admit decision | NOTIFY | NEW | 4 |
| Live ED board / triage / disposition | ACT | NEW | 4+ |
| ED flow/wait/resource/acuity/arrival analytics | WEB | MOCK | — |

## Transport · EVS · Staffing (mobile workers)

| Capability | Treatment | Backend | Phase |
|---|---|---|---|
| Transport request + 17-status lifecycle; **claim/progress/assign** | GLANCE·ACT·NOTIFY | LIVE | 1 |
| Transport **structured handoff** (to/summary/docs/risks) | ACT·NOTIFY | LIVE | 1 |
| Regional / inter-facility transfer | GLANCE·ACT | LIVE | 3 |
| EVS bed-turn + 5-status; **claim/start/complete**; isolation SOP | GLANCE·ACT·NOTIFY | LIVE | 1 |
| Staffing plan (gap/min-safe) + request; **create/source/assign/fill** | GLANCE·ACT·NOTIFY | LIVE | 3 |

## Process Improvement · Patient Flow/FHIR/Integration · Platform

| Capability | Treatment | Backend | Phase |
|---|---|---|---|
| PDSA cycle (plan/do/study/act) — **own/advance**; barrier ack | GLANCE·ACT·NOTIFY | NEW | 4 |
| Bottleneck / root-cause / process mining / 4D navigator | WEB | MOCK | — |
| Flow encounter/event/occupancy; patient identity; FHIR; ambient | feeds census/board | LIVE | 1–2 |
| Integration source/health/connector; dead-letter | GLANCE·NOTIFY (admin) | LIVE | 4 |
| Facility blueprint / space modeling | WEB | LIVE | — |
| Auth · biometric · workflow switcher · profile/prefs · search · directory | ACT·GLANCE | LIVE + NEW | 0–2 |
| Admin (users, auth providers) | WEB | LIVE | — |

**Coverage assertion** Every audited model maps to a row above or its domain’s WEB bucket.

| Capability | Treatment | Backend | Phase |
|---|---|---|---|

> The domains carrying the most **NEW** backend (ED actions, PI/PDSA, RTDC huddle action-items & discharge-readiness) are precisely those that are *mock/stub on the web today* — net-new product, not a mobile gap — which is why they land in P3–P4.

# Appendix B — Backend change log

Consolidated additive backend work (detail: `reference/04-backend-requirements.md`):

| Area | Work | Phase |
|---|---|---|
| **Auth** | `HasApiTokens`; `/auth/token{,/refresh,/revoke}`, `/auth/change-password` (honor `must_change_password`); OIDC+PKCE path; token abilities | 0 |
| **Push** | `mobile_devices` registry; APNs `.p8` + FCM v1 `PushNotifier`; notification router | 0→1 |
| **BFF** | `/api/mobile/v1/*` scaffold; per-domain endpoints rolled out P1–P4; uniform envelope; PHI-minimization | 0–4 |
| **Assignment/ RBAC** | `user_id` FKs on owners + `user_unit` pivot; Laravel Policies; remediate auth-less endpoints | 0 |
| **Broadcasts** | `BarrierUpdated` (P1); `DiversionUpdated` (P3); WS config endpoint | 1–3 |
| **Evaluators** | SLA / capacity deficit / barrier aging / unplaced bed (P1); ED boarding/LWBS/surge, staffing min-safe (P3); boarder dwell, integration health (P4) | 1–4 |
| **Bug fixes** | OR `store status_id`; analytics `actual_start/end_time` → orlog/metrics; reference `active_status` | before 2 |
| **Net-new** | huddle action-items API (P2); PI/PDSA write paths, ED write path, simulation read (P4) | 2–4 |

# Appendix C — Risk register

| Risk | Sev | Mitigation |
|---|---|---|
| **KMP commitment decided late** → costly re-architecture | High | Decide **now** (§5). Floor: separate apps + mandated OpenAPI contract + shared conformance spec |
| **Earned-urgency mis-tuned** → alarm fatigue or missed breach (safety) | High | Treat taxonomy as clinical-safety; configurable bands; clinical sign-off gate; track T1 precision + per-shift budget |
| **PHI leak** (notifications/logs/snapshots/SDKs) | High | PHI-free push by design + CI payload linter + FLAG_SECURE/privacy view + BAA/PHI-free SDKs; PHI-leak audit gate |
| **Auth additivity breaks the locked flow** | High | Strict additive design; auth regression suite as a P0 gate; never touch protected files |
| **Net-new backend gates ED/PI** slips features | Med | Sequence last (P4); ship glanceable signals first (P3); deep-link-to-web fallback |
| **Push reliability** on battery-constrained devices | Med | Push-first + delta-sync; never hold a bg socket; poll-on-resume fallback; real-device testing |
| **Apple Critical Alerts / Android FSI permission delays** | Med | Apply early in P2; degrade T1 to high-importance until granted |
| **Web endpoints currently auth-less** = security finding | Med | Remediate in P0; BFF is the only mobile surface |
| **Reference/OR backend bugs** block P2 | Low | Fix list scheduled before P2 (Appendix B) |

# Appendix D — Team & timeline

| Role | Count | Focus |
|---|---|---|
| **iOS (Swift/SwiftUI)** | 2 | native UI, widgets/Live Activities, watch |

| Role | Count | Focus |
|---|---|---|
| **Android (Kotlin/Compose)** | 2 | native UI, Glance widgets, Wear |
| **KMP / shared** | 1 | domain/data/sync/tokens (pairs with both) |
| **Backend (Laravel)** | 1–2 | BFF, auth, push, evaluators, RBAC, broadcasts |
| **Design** | 1 | token parity, mobile component specs, glanceability |
| **QA / security** | 1 | conformance + security/PHI gates |
| **Clinical-safety advisor** | part | notification taxonomy + tier sign-off |
| **PM / EM** | 1 | sequencing, store, rollout |

**Timeline:** ~9–12 months to GA; internal pilots from end of P1; P1–P2 overlap once the BFF pattern is set.

## Appendix E — Program definition of done

- The parity matrix is fully reconciled: every GLANCE/ACT/NOTIFY row shipped or explicitly deferred with a tracked backend ticket; every WEB row has a working deep link.
- **All seven security release-gate criteria** pass (`reference/06-security-hipaa.md §9`), notably: **zero PHI** leaves the boundary (notifications/logs/snapshots/analytics); **auth additivity** proven by a green regression suite over the locked flow; **biometric + auto-lock + remote-revoke**; **cert pinning** with rotation; **action RBAC** unbeatable by a tampered client; **audit trail** on every mutation; **named security + clinical-safety sign-off**.
- The notification taxonomy meets its precision/budget criteria in pilot and has clinical-safety sign-off.
- One design system, three platforms — verified non-divergent by the token CI check.

## Document map

This consolidated plan is the front door. Detail lives in:

### Deep-dive planning docs

- `00` Vision/Scope/Personas · `01` Feature-Parity Matrix · `02` Functionality by Role · `03` Architecture
- `04` Backend & API · `05` Notifications · `06` Security/HIPAA · `07` Roadmap & Phasing

### Scaffolding (begun)

- `design-tokens/` — DTCG source + Style Dictionary + samples.
- `api-contract/hummingbird-bff.v1.yaml` — validated OpenAPI 3.1.

### Evidence — the seven-stream code audit

`research/` (`01`–`07`: ED, RTDC, Perioperative, Improvement/Ops, Platform, Supporting domains, Mobile best practices).